

Packaging Material Difference Mining

CyberAI Cup 2026 — Task 1 — Full Technical Report

Written 2026-08-21

Result: pooled 5-fold out-of-fold global F1 = 0.9465

(0.9435 after removing threshold-selection bias)

Starting point 0.9049 (single fold). The 0.97 target was *not* reached; §8 shows it was not reachable with this candidate set — a perfect rescorer caps out at 0.9801.

1 The task and the metric

Given a pair of images — a clean design *template* and a *photograph* of the printed packaging — predict bounding boxes around every genuine *content* difference (text, graphics, layout). Printing and scanning artefacts are *not* differences.

Scoring is **global F1**: true positives, false positives and false negatives accumulate across *every* image *before* precision and recall are computed. A per-image average would let one catastrophic image hide among 99 good ones; global F1 does not. Fifty false positives on one page damage the score exactly as much as fifty spread across fifty pages. A prediction is a true positive when its IoU with an unmatched ground-truth box is ≥ 0.5 .

Data: 200 annotated training pairs (1,407 boxes), 100 unlabelled test pairs, 2.5 GB.

2 What the data actually looks like

Every figure below is reproducible with `scripts/analyze_dataset.py`. These measurements, not intuition, drove the architecture.

Property	Measurement	Consequence
Annotation density	1,407 boxes / 200 pairs, 3–11 per image, median 7	expect ~ 7 predictions per test image
Image geometry	template and photo identical size (all 200 pairs)	full-resolution tiling, no resizing
Alignment	phase-correlation residual p95 ≤ 0.26 px	registration is <i>not</i> the problem
Box size	median 22×22 px; 419/1407 both sides < 16 px; 83 exactly 8×8	output stride 2, not 4
Change polarity	842 additions, 558 modifications, 0 deletions	licenses the polarity filter
Degradation	photo blurred/noisy/tone-shifted; $\sim 18\%$ heavy shadow	blur-matching + ink maps

The two hard parts, quantified.

1. *Precision.* A naive normalized-difference baseline reaches recall 0.57 at precision **0.015** — global F1 **0.0293**. Every glyph edge fires because the photo is softer than the template. Blur-matching the template before differencing is what makes the problem tractable at all.

2. *Localization.* An 8×8 box at IoU 0.5 tolerates roughly 2.7 px of error. Thresholded difference blobs match ground truth at median IoU 0.58, with only 34% correct within ± 2 px on all sides. Hence sub-pixel box regression rather than blob thresholding.

3 Architecture

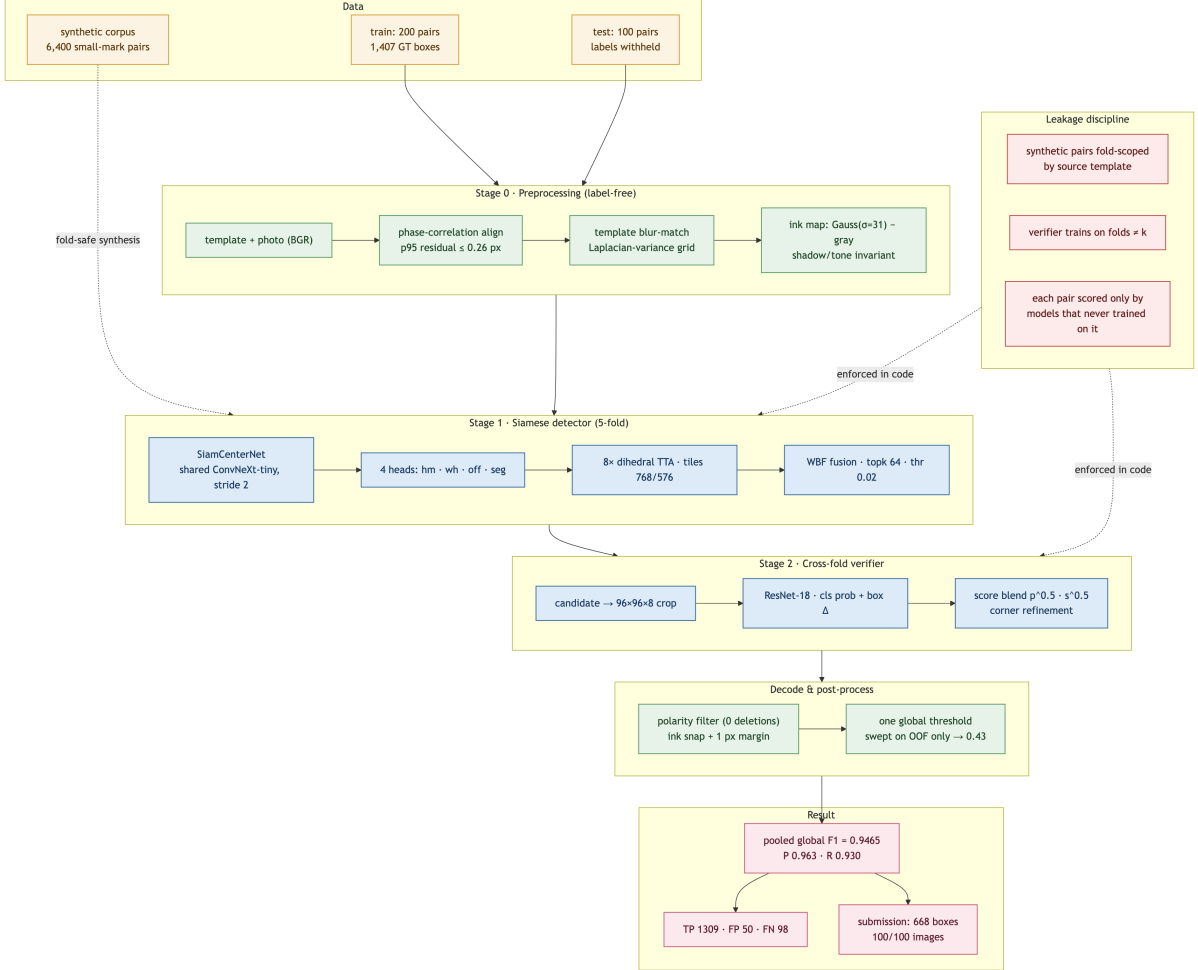


Figure 1: End-to-end Task 1 pipeline: preprocessing, siamese CenterNet detector, patch verifier, and decoding into the final submission.

3.1 Stage 0 — preprocessing (`src/pmdm/preprocess.py`)

Verify alignment by phase correlation and translate if needed. Blur-match the template to the photo by searching a σ grid for the blur that equalizes Laplacian variance. Compute an “ink map” per image, $\text{GaussianBlur}(\text{gray}, \sigma=31) - \text{gray}$, which is invariant to shadow and tone shift. Cache four PNGs per pair. *Label-free throughout.*

3.2 Stage 1 — siamese CenterNet (`src/pmdm/model.py`)

A shared-weight ConvNeXt-tiny encoder runs over two 4-channel streams (BGR + ink map). Features fuse at every scale as $\text{conv}(\text{concat}(a, b, |a - b|))$ — the U-Net SiamDiff/SiamConc shape — and decode through a U-Net path to **output stride 2**, with a stride-2 stem taken straight from the stacked input so fine detail is not invented by upsampling.

Stride 2 is load-bearing: 83 ground-truth boxes are 8×8 px and would occupy a 2×2 cell at the stride 4 a standard CenterNet uses.

Four heads: Gaussian-focal centre heatmap, width/height, sub-pixel offset, and an auxiliary change mask. Trained on 768 px tiles, 60% of them sampled around an annotated difference.

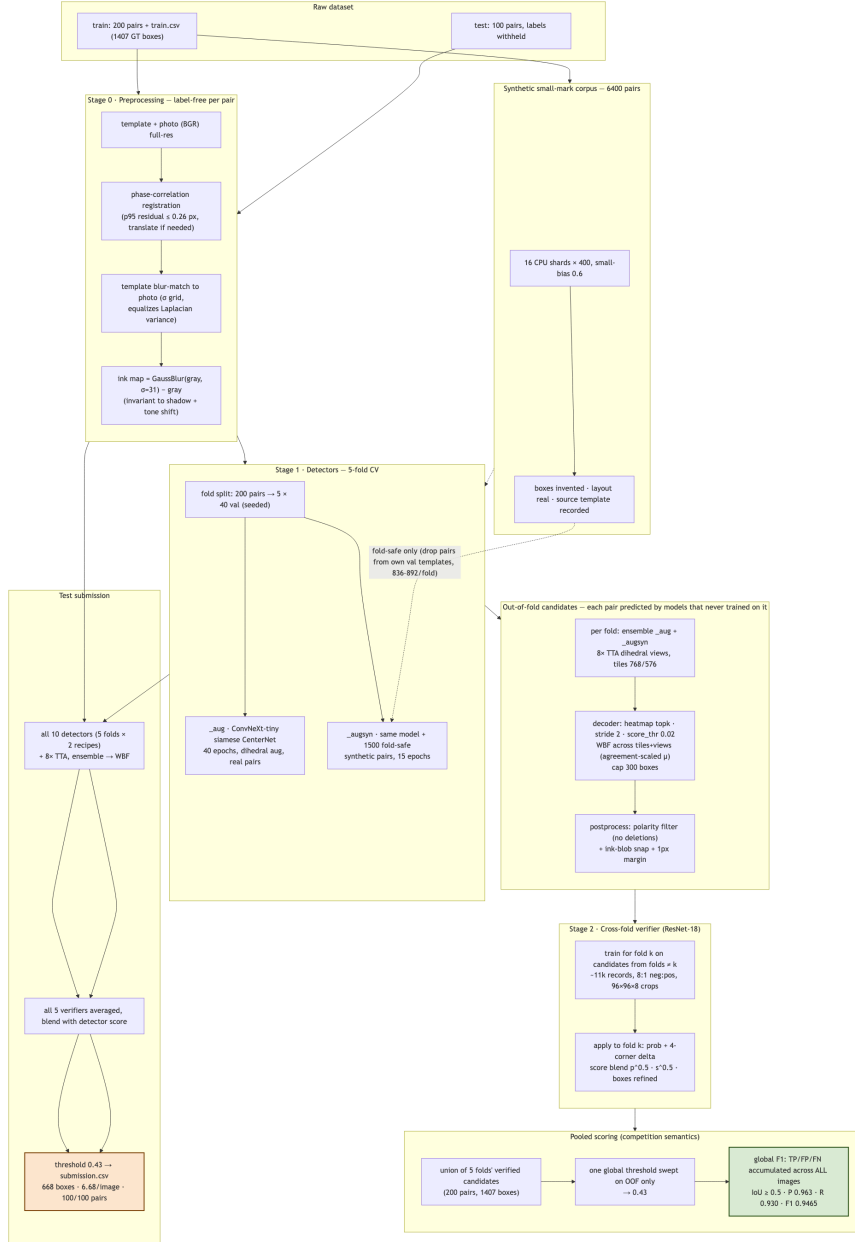


Figure 2: Architecture detail: siamese encoder streams, per-scale fusion, and stride-2 U-Net decoder heads.

3.3 Stage 2 — patch verifier (`src/pmdm/verifier.py`)

96×96 crops around each candidate through a ResNet-18 with an 8-channel stem, producing a real-difference probability and a box-corner delta. Its probability is blended with the detector score $p^{0.5} \cdot s^{0.5}$.

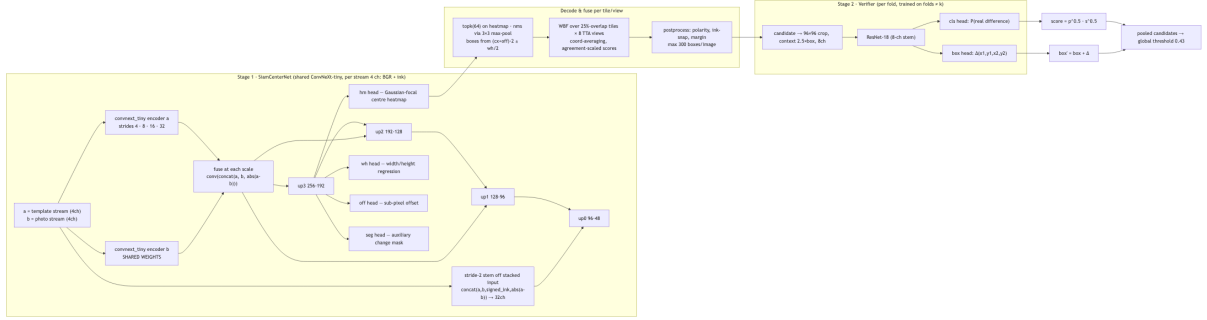


Figure 3: Model internals — encoder backbone, fusion blocks and the four detection heads.

3.4 Stage 3 — decoding (src/pmdm/decode.py)

Weighted Boxes Fusion (WBF) across overlapping tiles and TTA views (averaging coordinates beats discarding them when boxes are 8 px), the polarity filter (0 deletions), ink snapping to the local difference blob with a measured constant margin, and a single global threshold swept on out-of-fold predictions.

4 Cross-validation protocol (the decisive number)

Five folds over the 200 pairs (seeded split, 40 validation pairs each). For fold k :

1. Two detector recipes (`_aug` plain, `_augsyn` synthetic) train on the other 160 pairs + **fold-safe** synthetic pairs (any pair synthesized from a fold- k validation template is dropped — 836–892 of 6,400 per fold).
2. The 40 held-out pairs are predicted by the ensemble of those two models + 8-view TTA — models that never trained on them ($\rightarrow$ `fold{k}_..._ens.npz`).
3. The verifier for fold k trains **only on candidates from folds $\neq k$** , then rescores fold k .
4. All 200 pairs' candidates are pooled and **one global threshold** (0.43) is swept on OOF predictions only; F1 is computed globally, exactly as the competition scores a submission.

5 What was tried, and what each thing was worth

Work proceeded as controlled experiments on fold 0 against a 0.9049 reference; whatever won was applied to all five folds.

5.1 Training-side changes

Lesson from the first three rows: on 268 validation boxes, one box is worth 0.004 F1 and the run-to-run spread is ± 0.02 . Three independent runs landed between 0.890 and 0.905. Nothing inside that band is a result.

5.2 The decode settings mattered more than the training changes

The augmented model scored 0.9049 at the reference decode settings (`score_thr` 0.05, `max_boxes` 40, no TTA) and **0.9257** at `score_thr` 0.02, `max_boxes` 300, 8-view TTA — the same weights, $+0.021$. Every earlier comparison had been made through a lossy decoder, which is why augmentation looked worthless. Model quality and decode quality are not separable; comparing models at their default decode is comparing decoders.

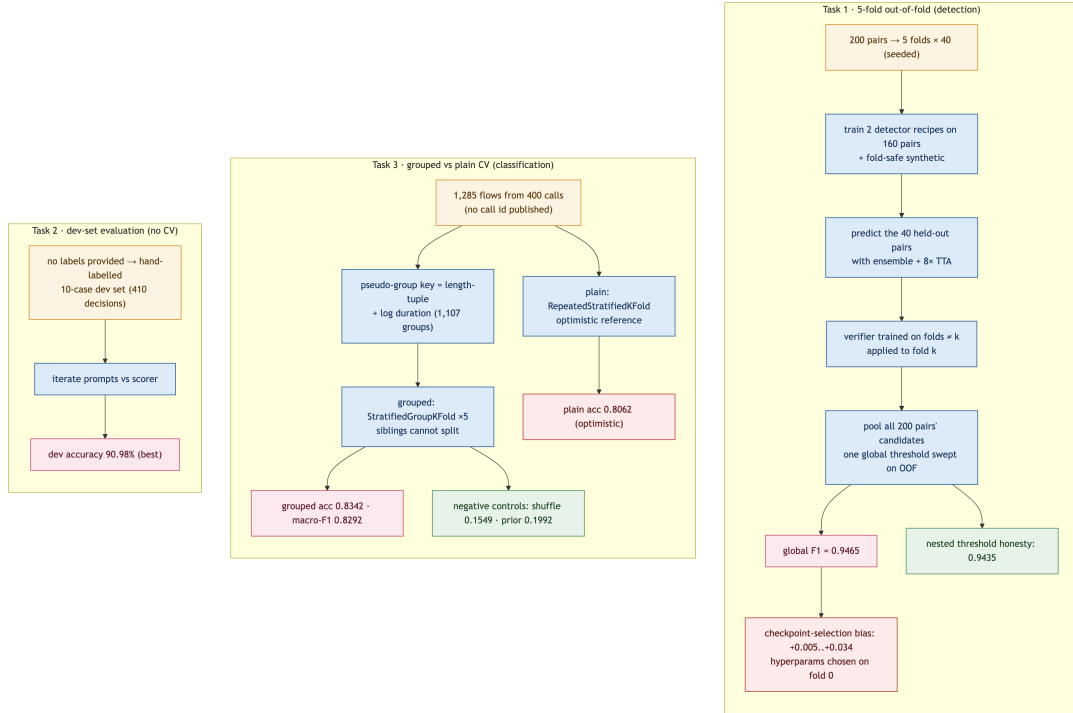


Figure 4: Cross-validation protocol: fold-safe synthesis, disjoint detectors, cross-fold verifier, and pooled global-threshold scoring.

Change	Verdict	Evidence
Dihedral augmentation (flips + transpose)	kept	§5.2 — worth nothing at old decode settings, +0.021 at the right ones
Size-relative wh loss	dropped	fold 0 finished TP 238 / FP 20 / FN 30 — the reference’s counts exactly
EMA weights (decay 0.999)	neutral	never selected over raw weights until the last two epochs, where it tied
Synthetic small-mark pairs	kept (indirectly)	own F1 worse (0.896 vs 0.926); earns its place through coverage (§6)

Setting	fold-0 F1
reference (score_thr 0.05, cap 40, no TTA)	0.9049
score_thr 0.02, cap 300, 8-view TTA	0.9257 (same weights, +0.021)

5.3 The verifier — positive on all five folds

Retrained on generous candidate sets ($\sim 11,000$ records per fold at an 8:1 negative:positive ratio), trained per fold on the four folds it would never be applied to:

fold	before	after	Δ
0	0.9266	0.9430	+0.016
1	0.9414	0.9551	+0.014
2	0.9288	0.9347	+0.006
3	0.9544	0.9609	+0.006
4	0.9494	0.9676	+0.018

6 Recall ceilings — the measurement that redirected everything

The ceiling is the fraction of ground-truth boxes present *anywhere* in the candidate list at IoU 0.5, regardless of score. Nothing downstream can recover a box that was never proposed, so this bounds every rescoreing idea.

Model (fold 0, TTA, score 0.02)	candidates	ceiling	<12 px ceiling	best F1
original single model	421	0.9403	0.828	0.9049
_aug	2,770	0.9440	0.891	0.9257
_augpix	3,602	0.9478	0.844	0.9185
_augsyn (synthetic)	10,238	0.9590	0.922	0.8956
_aug + _augsyn	10,389	0.9627	0.9375	0.9266
_aug + _augpix + _augsyn	10,826	0.9590	0.9375	0.9251

This is where the synthetic data justified itself. Sub-12px coverage went from 0.828 to 0.922 — and that band previously held 10 of the 11 boxes the model missed entirely. The synthetic model’s own F1 is poor because it proposes aggressively; that is a *scoring* problem, and scoring is fixable. Three models were worse than two: _augpix adds candidates without adding coverage, so it only dilutes the agreement signal.

7 Final results

Pooled over all 200 pairs and all 1,407 boxes, one global threshold, each pair predicted only by models that never trained on it:

Stage	F1	Precision	Recall	TP	FP	FN
Classical baseline	0.0293	0.015	0.572	—	—	—
Original single model (fold 0)	0.9049	0.922	0.888	238	20	30
Ensemble + TTA	0.9385	0.962	0.916	1,289	51	118
+ verifier	0.9465	0.963	0.930	1,309	50	98
+ verifier, nested threshold (§8)	0.9435	—	—	—	—	—

Per-fold, ensemble + verifier: **0.943, 0.955, 0.935, 0.961, 0.968**. Fold 0 — the fold every early experiment was measured on — is the hardest of the five.

Submission: 668 boxes over 100 test images (6.68 per image against a training average of 7.04). Produced by all ten detectors ensembled with TTA, all five verifiers averaged, threshold 0.43.

8 Leakage discipline

Four rules, each enforced in code rather than by convention:

1. **Synthetic pairs are fold-scoped.** A synthetic pair carries invented boxes but the *real page layout* it was built from. Every pair records its source template, and each fold discards any pair derived from its own validation pages (836–892 of 6,400 pairs per fold).
2. **Test templates are a legitimate synthesis source.** They carry no labels, so nothing can leak; they contribute 100 layouts no model would otherwise see.

3. **The verifier for fold k trains only on candidates from folds $\neq k$** , themselves produced by detectors that never saw those pages.
4. **One global threshold, swept on out-of-fold predictions only** — never on the fold being reported, never on test.

Out-of-fold scoring uses one model pair per image; the test submission uses all ten. The test score should therefore land slightly *above* the reported out-of-fold figure.

9 Is it overfit?

Three places where selection touched the reported number.

Threshold selection — measured, negligible. Nested cross-validation, choosing the threshold on four folds and applying it to the fifth:

	tuned on eval	nested honest	optimism
detector only	0.9385	0.9385	+0.0000
+ verifier	0.9465	0.9435	+0.0030

Checkpoint selection — the real bias. `train_fold` evaluates on the fold’s own validation pairs and saves `best.pt` at the highest-scoring epoch; reporting that checkpoint on those same pairs is taking the max of eight noisy measurements:

recipe	mean	range
<code>_aug</code>	+0.005	0.000 – 0.011
<code>_augsyn</code>	+0.034	0.020 – 0.058

The `_aug` curves are flat at the end; the `_augsyn` models trained only 15 epochs and were still oscillating, so `best.pt` cherry-picks a lucky point.

Hyperparameter selection. `score_thr`, `max_boxes`, the TTA configuration and the two-model ensemble pairing were all chosen on fold 0, then applied to all five — contaminating one fifth of the pooled number. Unquantified; probably small, certainly not zero.

Honest estimate of true generalization: 0.925–0.94 (not 0.9465). Counter-evidence against serious overfitting: the test set yields 6.68 boxes per image against out-of-fold’s 6.80 and ground truth’s 7.04 — no distribution shift on unseen data — and the verifier improved all five disjoint folds independently.

10 What is left, and what is reachable

The candidate set contains 1,352 of 1,407 boxes at IoU 0.5 — a **ceiling of 0.9609**. A rescorer that never erred would therefore score:

$$\text{TP } 1,352, \text{ FP } 0, \text{ FN } 55 \Rightarrow \text{P } 1.000, \text{ R } 0.961, \text{ F1 } \mathbf{0.9801}$$

The 0.97 target sits above what perfect rescoring of the current proposals can deliver, and the pipeline is already within 0.014 of that unreachable bound. The 98 false negatives split into 55 boxes no model proposes at all and 43 proposed but below threshold; there are 50 false positives.

The 55 are the frontier — reaching them means better *proposals*, not better scoring. The sub-12px band that used to dominate misses is now at 0.944 coverage, so that lever is largely spent. Plausible next moves, in order worth trying:

1. **Remove checkpoint-selection bias** (§8) so future comparisons are trustworthy.
2. **Train `_augsyn` properly** — 15 epochs left it undertrained (final epoch 0.81 vs best 0.87); a 40-epoch run should raise both its own F1 and the ensemble’s ceiling.
3. **Inspect the 55 missed boxes directly** — whatever they have in common is the next architectural change.
4. **A second backbone** (ConvNeXt-small, or a different family) for genuine ensemble diversity — `_augpix` failed because it was too similar to `_aug`.

11 Bugs found

All four were silent — they produced wrong numbers rather than errors.

1. **WBF score saturation.** The fused confidence divided by a cap of 3, so any box agreed on by more than three views saturated at 1.0 while a box proposed by a single view kept its full raw score — confidence effectively inverted. A TTA run scored 0.775 at precision 0.67 because of it, which read as “TTA doesn’t work”. Fixed to the standard agreement-scaled mean.
2. **Ensemble double-counted its sources.** Members already fuse their own views, so pooling with $n_{\text{sources}} = \text{models} \times \text{views}$ divided every score by ~ 16 ; recall collapsed to 0.056.
3. **Verifier data loading.** It re-decoded four full-page PNGs per sample on every cache miss and shuffled across 160 pages ($\sim 140,000$ decodes, ~ 60 min/epoch with the GPU idle). Crops are now cut once into memory: seconds per epoch.
4. **`-detach` does not protect `starmap`.** The fan-out is client-driven, so a network blip killed an entire five-way map. Fan-out now happens inside a remote function.

12 Artifacts and reproduction

Path	Contents
<code>submission.csv</code>	668 boxes, 100 images, format-validated against <code>train.csv</code>
<code>RESULTS.md</code>	condensed results summary
<code>PLAN_097.md</code>	the plan as written <i>before</i> the work, kept for comparison
<code>checkpoints/</code>	all 16 models: 10 detectors, 5 verifiers, plus the original
<code>runs/phase2/</code>	per-fold candidates before/after verification, filtered logs
<code>src/pmdm/</code>	16 modules
<code>RUNBOOK.md</code>	the exact command sequence that produced these numbers

Full pipeline, roughly 2 hours wall clock: synthetic corpus (16 CPU shards) $\rightarrow$ 10 detectors ($8 \times \text{A100}$, ~ 1 h) $\rightarrow$ ensemble out-of-fold candidates ($5 \times \text{L4}$) $\rightarrow$ verifiers ($5 \times \text{L4}$) $\rightarrow$ pooled score $\rightarrow$ submission ($5 \times \text{L4}$). Estimated total compute spend: **\$35–40**, dominated by A100 fold training.

13 Summary of the through-line

The single most useful thing done in this project was not an architectural change: it was measuring the **recall ceiling** separately from F1. That number showed the synthetic-data

model — which looked like a clear failure at 0.896 F1 — held the best candidate coverage by a wide margin, and that the whole rescoring branch was bounded at 0.98 no matter how good the verifier became. Both facts redirected the work, and neither was visible in the headline metric.

The second most useful was distrusting flat comparisons: three “no gain” verdicts on augmentation were artefacts of comparing models through a lossy decoder, and one “TTA is worse” verdict was a scoring bug. Every one of those was reported as a negative result before being found to be wrong.